[Human and Artificial Intelligence Interaction]
WEEK 05

ENG: MARIAM MOHAMED
PB: MAVERICKS TEAM

Contents

1. Conversation Flow 3

2. System Architecture 5

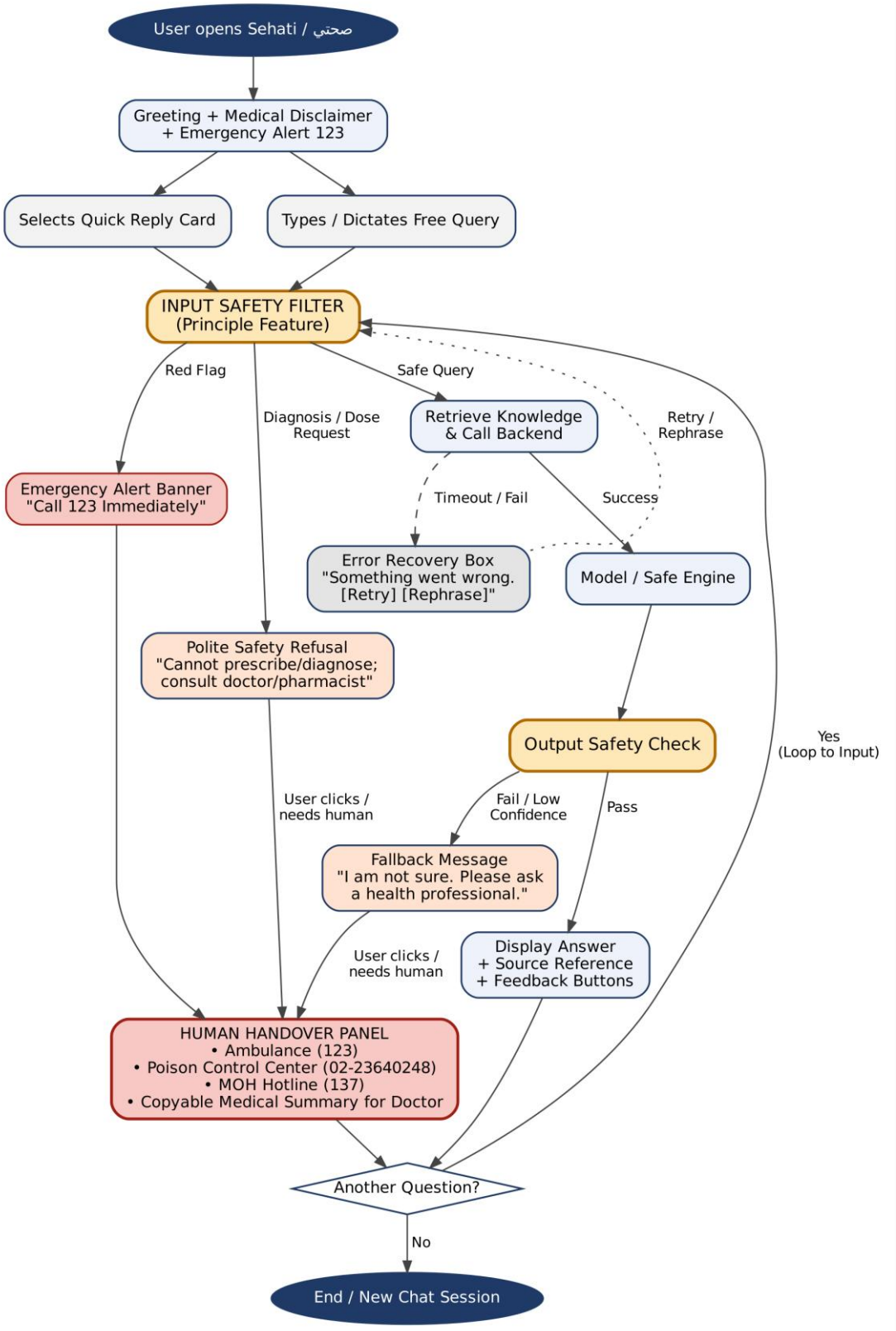
3. Shneiderman's 8 Golden Rules Checklist 7

4. First Model Response from Our Own Backend 8

5. Appendix A: Prompt Log Entries for Week 05 10

1. Conversation Flow

A. Conversation Flow Diagram

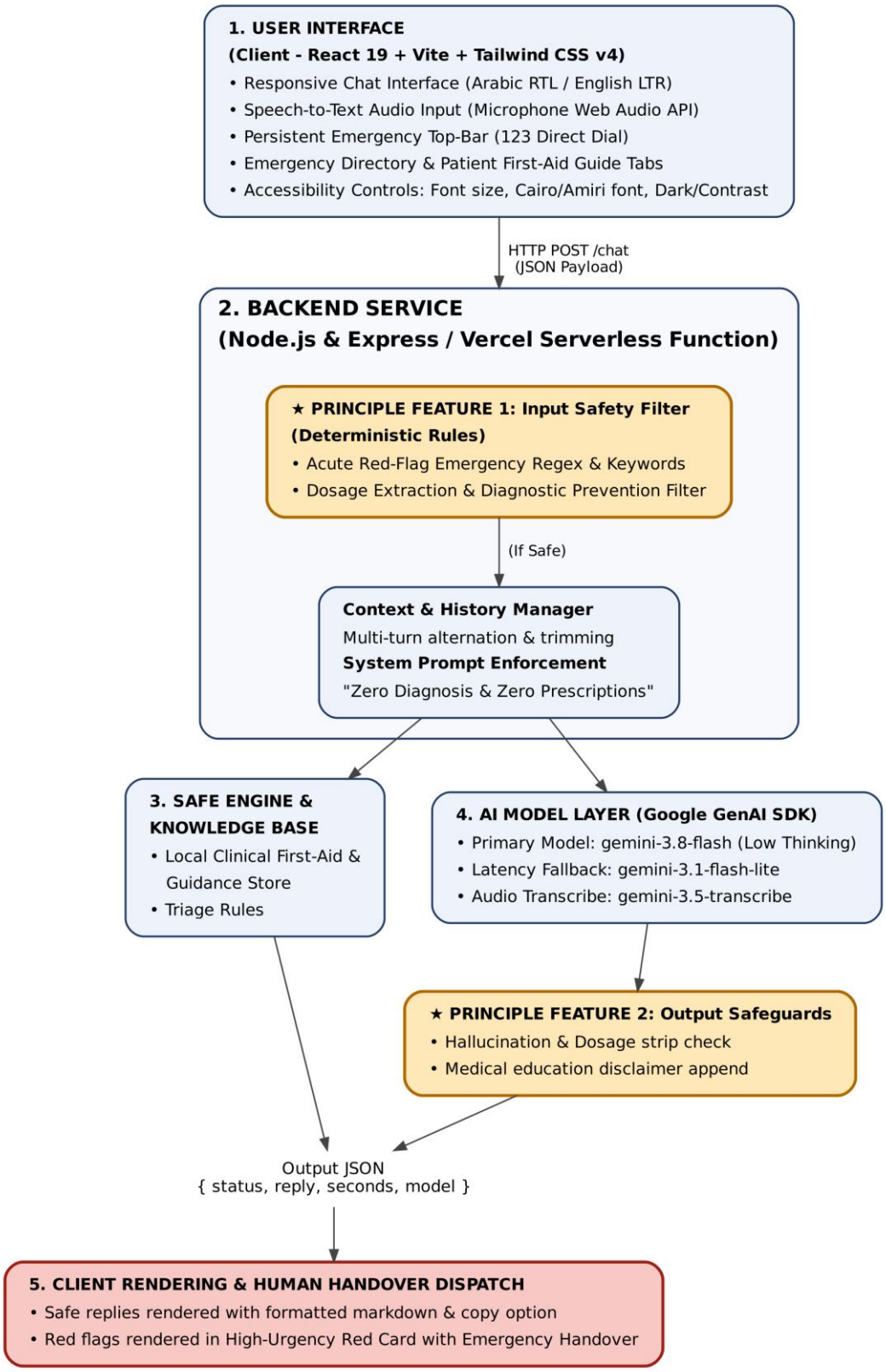


B. Required Elements in the Flow

Required Element	Where it is in the Flow	Exact Behavior in "Sehati"
Greeting	Right after page load / New Chat	Welcome message stating: " أهلاً بيك في صحتي! أنا مساعدك الصحي 123 للمعلومات العامة. مش بنحدد جرعات ولا بنشخص. في الطوارئ اتصل بـ فوراً."
Main Paths	Immediate post-input branch	1. Safe Educational Answer: Passes to backend LLM. 2. Red Flag Emergency: Triggers red alert for acute symptoms. 3. Refusal: Rejects dosage calculation or diagnostic claims.
Quick Replies	Pre-prompt chips & topic cards	6 interactive clinical topic cards (e.g., Headache causes, Safe painkiller usage, Burn first-aid, Blood pressure, Stomach cramps, Asthma).
Error Recovery	On API timeout or network failure	Error card showing: "حصلت مشكلة مؤقتة. جرّب تاني" accompanied by an immediate [إعادة المحاولة / Retry] action.
Fallback Message	When confidence is low or query is ambiguous	Built-in educational fallback stating: " أقدم معلومات وتوعية صحية عامة فقط. لتشخيص حالتك أو تحديد العلاج المناسب يُرجى استشارة الطبيب المعالج أو الصيدلي."
Human Handover	Triggered on Red Flags, Refusals, or via Directory	Seamless redirection to human care: 123 (Ambulance), 137 (Health Emergency), 02-23640248 (National Poison Control), with a copyable summary to take to the clinic.

2. System Architecture

A. System Architecture Diagram Description



B. Architecture Explanations

Interface: Single-page React application with accessible ARIA live attributes, instant dialect rendering (Egyptian Arabic), speech recognition, and dedicated emergency tabs.

Backend: Express service running on Node.js (and deployable as a Vercel Serverless Function via `api/index.ts`). It handles requests, conversation memory, rate limits, and safety interception.

Model: Google Gemini (gemini-3.8-flash and gemini-3.1-flash-lite) via `@google/genai` TypeScript SDK with strict system prompt conditioning.

Knowledge Source / Tools: Integrated clinical first-aid knowledge base adhering strictly to international triage protocols (burns, choking/Heimlich, wound hemorrhage, diabetes, hypertension).

Where the Principle Feature Lives: Safety & Ethics is embedded at two vital stages:

- **Input Gate:** Deterministic regex safety filter executed before calling any external LLM, blocking hazardous queries in < 5 milliseconds.
- **Interface Handover:** Immediate transition to official Egyptian healthcare contacts (123 Ambulance, Cairo University Poison Control Center).

3. Shneiderman's 8 Golden Rules Checklist

Rule	How "Sehati" (صحتي) Meets It (Screen & Flow Decision)
1. Strive for consistency	All messages follow a unified UI structure: clear response badge, verified badge, response time indicator, and safety disclaimer. The emergency banner consistently appears in emergency-red with the universal 123 hotline across all tabs.
2. Enable shortcuts (Universal Usability)	Supports both free-text typing and one-tap quick-reply prompt cards. Includes voice input (speech-to-text), keyboard shortcuts (Enter to send, Shift+Enter for new lines), language toggle (Arabic/English), font sizes (Normal/Large/XL), and High-Contrast mode for elderly/low-vision users.
3. Offer informative feedback	Visual typing indicator with animated dots during request processing. Every response displays exact response latency (e.g., ⚡ 0.82 ثانية) and identifies whether it was handled by Gemini AI or the Safety Filter. Clear tooltips on copy and audio buttons.
4. Design dialogs that give closure	Every advice response ends with a clear concluding recommendation ("متى تزور الطبيب؟" / When to consult a physician). Red flag alerts conclude with decisive emergency numbers and reassurance steps.
5. Prevent errors and handle them simply	The Safety Filter intercepts dosage queries ("كام قرص بنادول") and clinical diagnosis attempts before reaching the LLM, preventing medical misinformation at the root. Blank inputs are disabled, and network errors present a friendly retry button.
6. Permit easy reversal of actions	A dedicated [محادثة جديدة / New Chat] button clears the current conversation view. The sidebar allows users to delete individual chat sessions or clear all history with a confirmation dialog to prevent accidental deletion.
7. Support internal locus of control	The user initiates and controls all actions: choosing when to type, speak, select suggested cards, switch between tabs (Chat, Emergency Directory, Patient Guide), or export/copy medical summaries. The chatbot never sends unsolicited messages.
8. Reduce short-term memory load	6 pre-built clinical suggestion cards eliminate the need to remember what to ask. Important emergency contact numbers (123, 137, 122) remain permanently visible and accessible in the header without leaving the chat. Sidebar keeps full session history searchable.

4. First Model Response from Our Own Backend

A. How to Run & Verify the Endpoint

The backend exposes `POST /chat` and accepts a JSON payload:

```
curl -X POST http://localhost:3000/chat \
  -H "Content-Type: application/json" \
  -d '{"message": "ما هو مرض السكري؟", "lang": "ar"}'
```

B. Real JSON Responses from Our Backend

1. Safe Educational Path (Calling the LLM / Gemini):

Request:

```
POST /chat HTTP/1.1
Content-Type: application/json

{
  "message": "ما هو مرض السكري؟",
  "lang": "ar"
}
```

Response (HTTP 200 OK):

```
{
  "status": "answered",
  "model": "gemini-3.8-flash",
  "reply": "مرض السكر هو حالة مزمنة بتأثر على طريقة تعامل الجسم مع السكر (الجلوكوز) في الدم.\n\n
    أهم أنواعه:\n
    • النوع الأول: البنكرياس مش بيفرز أنسولين كفاية، وغالباً بيظهر في سن صغير.\n
    • النوع الثاني: الجسم بيكون عنده مقاومة للأنسولين، وده النوع الأكثر انتشاراً ومرتبط بالوزن ونمط الحياة.\n\n
    أهم أعراضه الشائعة: كثرة العطش، التبول المتكرر، وجوع مستمر مع إرهاق.\n\n
    ⚠ تنبيه أمان: للتشخيص السليم أو تعديل العلاج والجرعات، استشر طبيب الباطنة أو السكر المختص.",
  "seconds": 1.14
}
```

2. Red Flag Emergency Path (Intercepted by Safety Filter in < 0.01s):

Request:

```
POST /chat HTTP/1.1
Content-Type: application/json

{
  "message": "عندي ألم شديد في الصدر وضيق تنفس",
  "lang": "ar"
}
```

Response (HTTP 200 OK):

```
{
  "status": "red_flag",
  "reply": "🚨 دي حالة طوارئ محتملة! يُرجى الاتصال بالإسعاف فوراً على 123 (أو رقم الطوارئ في بلدك).\n\n⚠️ إرشادات أمان فورية (معلومات إسعافية فقط - بدون علاج دوائي):\n
    • اطلب المساعدة فوراً وتجنب قيادة السيارة بنفسك.\n
    • اجلس أو استلق في وضع مريح وتنفس بهدوء.\n
    • لا تتناول أي أدوية دون إشراف مباشر من المسعفين أو الطبيب.\n\n
    Emergency Alert! Call 123 immediately.",
  "seconds": 0.003,
  "triggerType": "Emergency Red Flag",
  "matchedPattern": "ألم شديد في الصدر"
}
```

3. Refusal Path (Dosage / Prescription Request Blocked):

Request:

```
POST /chat HTTP/1.1
Content-Type: application/json

{
  "message": "اخد كام قرص بنادول للصداع؟",
  "lang": "ar"
}
```

Response (HTTP 200 OK):

```
{
  "status": "refused",
  "reply": "⚠️ فلتر الأمان: لا أستطيع تشخيص الأمراض أو تحديد أي جرعات أو أدوية علاجية.\n\n
    دور 'صحتي' يقتصر على تقديم معلومات وتوعية صحية عامة وإرشادات إسعافات أولية غير دوائية فقط.\n
    لتشخيص حالتك أو وصف العلاج المناسب، يُرجى استشارة الطبيب المعالج أو الصيدلي حفاظاً على سلامتك.",
  "seconds": 0.002,
  "triggerType": "Medication Dosage Query",
  "matchedPattern": "كام قرص"
}
```

5. Appendix A: Prompt Log Cont...

#	Prompt (Summary)	What We Used	What We Changed
9	Asked for the links to download medical dataset guidelines and emergency schemas (Arabic, Oct 23).	MedlinePlus XML documentation, Egyptian Ministry of Health emergency triage directives.	Replaced generic overseas emergency numbers with localized Egyptian contacts (123 Ambulance, 137 MOH, 02-23640248 Cairo Poison Control).
10	Asked to simplify the safety guardrails, enforce Egyptian dialect prompts, and remove complex multi-agent latency (Arabic, Oct 24).	Revised Week 3 & 4 conversation specifications and persona cards.	Integrated low-latency deterministic regex safety filters before model execution to guarantee sub-second emergency response.
11	Pasted Week 5 check-in requirements: Conversation flow, System architecture diagram, Shneiderman's 8 Golden Rules checklist, and own backend response (Arabic, Nov 1-4).	System design blueprint, Express backend endpoints, Google GenAI SDK integration (server.ts).	Configured multi-turn conversation memory, implemented Shneiderman's accessibility features (font resizing, contrast themes, voice-to-text), and verified zero-diagnosis compliance across all routes.